CO1_AT2_Application based questions

Question 1: Hospital Medicine Distribution (10 Marks)

A hospital pharmacy stores medicine details in JSON format and daily medicine issue records in CSV format. Write a Python program to integrate the datasets, identify medicines nearing stock depletion, and generate a replenishment report.

Code:

```
import json
import csv

# Read JSON file
with open("medicine.json", "r") as file:
    medicines = json.load(file)

# Read CSV file
issue_data = {}

with open("issue.csv", "r") as file:
    reader = csv.DictReader(file)
    for row in reader:
        issue_data[row["MedicineID"]] = int(row["Issued"])

print("Replenishment Report")
print("-" * 50)

for med in medicines:
    issued = issue_data.get(med["MedicineID"], 0)
    remaining = med["Stock"] - issued

    print("Medicine:", med["Medicine"])
    print("Remaining Stock:", remaining)

    if remaining <= med["Threshold"]:
        print("Status: Reorder Required")
```

```
    else:
        print("Status: Stock Available")

    print("-" * 50)
```

Output:

```
Replenishment Report
--------------------------------------------------
Medicine: Medicine_1
Remaining Stock: 759
Status: Stock Available
--------------------------------------------------
Medicine: Medicine_2
Remaining Stock: 639
Status: Stock Available
--------------------------------------------------
Medicine: Medicine_3
Remaining Stock: 306
Status: Stock Available
--------------------------------------------------
Medicine: Medicine_4
Remaining Stock: 733
Status: Stock Available
--------------------------------------------------
Medicine: Medicine_5
Remaining Stock: 702
Status: Stock Available
--------------------------------------------------
Medicine: Medicine_6
Remaining Stock: 105
Status: Stock Available
--------------------------------------------------
Medicine: Medicine_7
Remaining Stock: 60
Status: Reorder Required
```

Question 2: Online Learning Platform (10 Marks)

An online learning platform stores course information in XML format and student enrollment data in CSV format. Write a Python program to merge the datasets, calculate the enrollment count for each course, and generate a popularity report.

Code:

```
import xml.etree.ElementTree as ET
import csv

# Read course details from XML file
tree = ET.parse("courses.xml")
root = tree.getroot()

courses = {}

for course in root.findall("Course"):

    course_id = course.find("CourseID").text
    course_name = course.find("CourseName").text

    courses[course_id] = course_name

# Read enrollment data from CSV file
enrollment_count = {}

with open("enrollments.csv", "r") as file:

    reader = csv.DictReader(file)

    for row in reader:

        course_id = row["CourseID"]

        if course_id in enrollment_count:
            enrollment_count[course_id] += 1
        else:
            enrollment_count[course_id] = 1
```

```
# Generate Popularity Report
print("\nCOURSE POPULARITY REPORT")
print("-" * 50)

for course_id, course_name in courses.items():

    total = enrollment_count.get(course_id, 0)

    print("Course ID        :", course_id)
    print("Course Name      :", course_name)
    print("Enrollment Count :", total)

    print("-" * 50)
```

Output:

```
COURSE POPULARITY REPORT
--------------------------------------------------
Course ID        : C001
Course Name      : Course_1
Enrollment Count : 11
--------------------------------------------------
Course ID        : C002
Course Name      : Course_2
Enrollment Count : 9
--------------------------------------------------
Course ID        : C003
Course Name      : Course_3
Enrollment Count : 12
--------------------------------------------------
Course ID        : C004
Course Name      : Course_4
Enrollment Count : 9
--------------------------------------------------
Course ID        : C005
Course Name      : Course_5
Enrollment Count : 11
--------------------------------------------------
Course ID        : C006
Course Name      : Course_6
Enrollment Count : 13
--------------------------------------------------
Course ID        : C007
Course Name      : Course_7
Enrollment Count : 15
--------------------------------------------------
```